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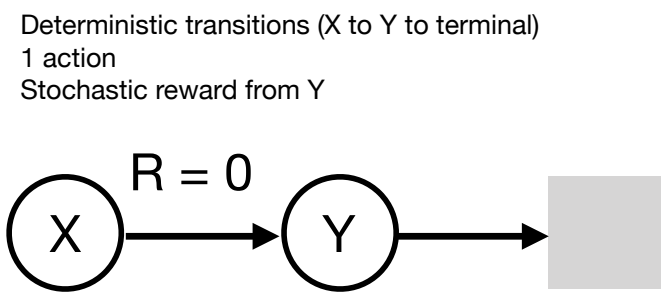
CMPUT 365: Introduction to Reinforcement Learning

Worksheet 6: Temporal Difference Learning Methods for Prediction

Question 1. Assume the agent interacts with a simple two-state MDP shown below. Every episode begins in state X, and ends when the agent transitions from state Y to the terminal state (denoted by gray box). Let's denote the set of states as $\mathcal{S} = \{X, Y, \top\}$. There is only one possible action in each state, so there is only one possible policy in this MDP. Let's denote the set of actions $\mathcal{A} = \{A\}$. In state Y the agent terminates when it takes action A and sometimes gets a reward of +1000, and sometimes gets a reward of -1000: the reward on this last transition is stochastic. Let $\gamma = 1.0$.



$$P(R = r|Y) = \begin{cases} 0.5 & \text{if } r = -1000 \\ 0.5 & \text{if } r = +1000 \end{cases}$$

(a) Write down $\pi(a|s)\ \forall\ s \in \mathcal{S}, a \in \mathcal{A}$.

(b) Write down all the possible trajectories (sequence of states, actions, and rewards) in this MDP that start from state X.

(c) What is the value of policy π (i.e. what is $v_\pi(X), v_\pi(Y)$)?

(d) Assume our estimate is equal to the value of π. That is $V(s) = v_\pi(s)\ \forall\ s \in \mathcal{S}$. Now compute the TD-error $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$ for the transition from state Y to the terminal state, assuming $R_{t+1} = +1000$. Why is the TD-error not zero if we start with $V(Y) = v_\pi(Y)$?

(e) Based on your answer to (d), what does this mean for the TD-update, for constant $\alpha = 0.1$? Will $V(Y) = v_\pi(Y) = 0$ after we update the value? Recall the TD-update is $V(S_t) \leftarrow V(S_t) + \alpha\delta_t$. What does this tell us about the updates TD(0) would make on this MDP?

(f) What is the expected TD-update, from state Y for the given V?

(g) Assume still that $V = V_\pi = 0$. What is the expectation and the variance of the TD update from state X? What is the expectation and the variance of the Monte-carlo update from state X?

Question 2. Suppose that in an environment, state transitions are deterministic and that reward is bounded so that $R_{min} = 0$ and $R_{max} = 1$, with $\mathbb{E}[R_t] = 0.5$. Find the maximum and minimum possible TD error $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$, where $\gamma = 0.9$ and $V = v_\pi$ with deterministic policy π.

Question 3. Modify the Tabular TD(0) algorithm for estimating v_π, to estimate q_π.

Tabular TD(0) for estimating v_π

Input: the policy π to be evaluated
Algorithm parameter: step size $\alpha \in (0, 1]$
Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(\mathit{terminal}) = 0$

Loop for each episode:
  Initialize S
  Loop for each step of episode:
    $A \leftarrow$ action given by π for S
    Take action A, observe R, S'
    $V(S) \leftarrow V(S) + \alpha\big[R + \gamma V(S') - V(S)\big]$
    $S \leftarrow S'$
  until S is terminal

Question 4. [**Challenge**] Exercise 6.1 from Sutton & Barto (2018).

Question 5. Exercise 6.2 from Sutton & Barto (2018).

Question 6. Exercise 6.3 from Sutton & Barto (2018).

Question 7. Exercise 6.4 from Sutton & Barto (2018).

Question 8. Exercise 6.5 from Sutton & Barto (2018).

Question 9. Exercise 6.6 from Sutton & Barto (2018).

Question 10. Exercise 6.7 from Sutton & Barto (2018).